MI4 Review: Vectors

- **What is a Vector? What is a Scalar?**

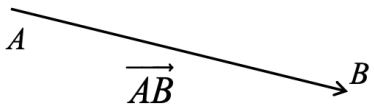
  A vector **v** is a quantity that has a length (or magnitude) and a direction.

  A scalar is a real number.

- Notation:

  We can either write the variable $\vec{v}$ with an arrow or using boldface **v**.

  We write $\overrightarrow{AB}$ for the vector with initial point (tail) A and terminal point (head) B.

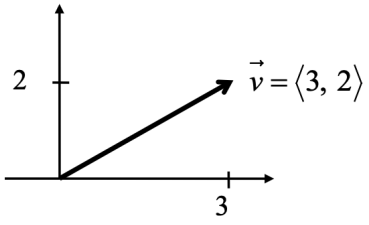


- **Vector Components**:

  A vector can be written in terms of its x- and y-components.

  Notation: We can either write $\mathbf{v} = \langle a, b \rangle$ or $\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$ or $\mathbf{v} = a\mathbf{i} + b\mathbf{j}$.

  Example:
  $\mathbf{v} = \langle 3, 2 \rangle = \begin{bmatrix} 3 \\ 2 \end{bmatrix} = 3\mathbf{i} + 2\mathbf{j}$



- **Magnitude**: $|\langle a, b \rangle| = \sqrt{a^2 + b^2}$

  Example: $|\mathbf{v}| = |\langle 3, 2 \rangle| = \sqrt{3^2 + 2^2} = \sqrt{13}$

- $\mathbf{v} = \langle |\mathbf{v}| \cos(\theta), |\mathbf{v}| \sin(\theta) \rangle$, where θ is the angle from the positive x-axis to the vector.

- **Vector Operations**:

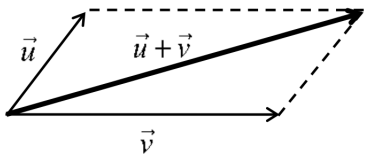
  1. Vector Addition: $\langle a_1, b_1 \rangle + \langle a_2, b_2 \rangle = \langle a_1 + a_2, b_1 + b_2 \rangle$ (The sum of two vectors is a vector.)
  2. Vector Subtraction: $\langle a_1, b_1 \rangle - \langle a_2, b_2 \rangle = \langle a_1 - a_2, b_1 - b_2 \rangle$ (The difference of two vectors is a vector.)
  3. Scalar Multiplication: $c\langle a, b \rangle = \langle ca, cb \rangle$ (The product of a scalar and a vector is a vector.)
  4. Dot Product: $\langle a_1, b_1 \rangle \cdot \langle a_2, b_2 \rangle = a_1 a_2 + b_1 b_2$ (The dot product of two vectors is a scalar. The dot product is important because of the geometrical interpretation of dot product.)

- Examples:

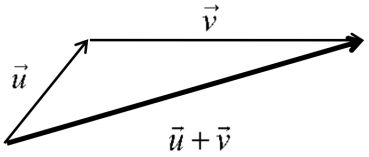
Vector Addition	Vector Subtraction	Scalar Multiplication	Dot Product
$\langle 3, 2 \rangle + \langle 4, 1 \rangle = \langle 7, 3 \rangle$	$\langle 3, 2 \rangle - \langle 4, 1 \rangle = \langle -1, 1 \rangle$	$4\langle 3, 2 \rangle = \langle 12, 8 \rangle$	$\langle 3, 2 \rangle \cdot \langle 4, 1 \rangle = 14$

- **Properties of Vector Operations**:
- Graphical method for adding vectors.

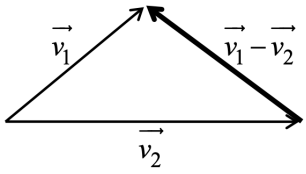
parallelogram method



head to tail (consecutive) method



- Graphical method for subtracting vectors.



- Scalar multiplication:

  If $c > 0$, then vectors $\mathbf{v}$ and $c\mathbf{v}$ point in the same direction.

  If $c < 0$, then vectors $\mathbf{v}$ and $c\mathbf{v}$ point in opposite directions.

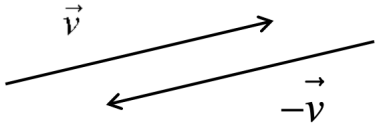
  Example: $\mathbf{v}$ and $-3\mathbf{v}$ point in opposite directions, and $-3\mathbf{v}$ is 3 times as long as $\mathbf{v}$.

- Additive inverse:

  The vector $-\mathbf{v} = -1 \cdot \mathbf{v}$ is the additive inverse of $\mathbf{v}$.

  $$\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$$

  $-\mathbf{v}$ has the same length as $\mathbf{v}$ but points in the opposite direction as $\mathbf{v}$.



- Distributive properties between vector addition and scalar multiplication:

$$c(\mathbf{u}+\mathbf{v}) = c\mathbf{u} + c\mathbf{v}$$
$$(a+b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$$

- **Unit Vectors**:

  Unit vector: A vector with magnitude equal to 1

  Standard unit vectors: $\mathbf{i} = \langle 1, 0 \rangle$ and $\mathbf{j} = \langle 0, 1 \rangle$

  All two-dimensional vectors can be written as a linear combination of the standard unit vectors $\mathbf{i}$ and $\mathbf{j}$.

  Example: $\mathbf{v} = \langle 5, -2 \rangle = 5\mathbf{i} - 2\mathbf{j}$

  A unit vector may be written as $\langle \cos\theta, \sin\theta \rangle$, for some angle θ.

  The unit vector in the direction of $\mathbf{u}$ is the vector $\hat{\mathbf{u}} = \dfrac{\mathbf{u}}{|\mathbf{u}|}$.

  Example: The unit vector in the direction of $\mathbf{v} = \langle 3, 2 \rangle$ is the vector $\hat{\mathbf{v}} = \left\langle \dfrac{3}{\sqrt{13}}, \dfrac{2}{\sqrt{13}} \right\rangle$.

Calculus with Vectors

- **Vector-Valued Functions**:

  A vector-valued function is a function of the form $\mathbf{r}(t) = \langle x(t), y(t) \rangle = x(t)\mathbf{i} + y(t)\mathbf{j}$.

  We can think of t as a parameter (often representing time).

  Examples:

$$\mathbf{r}(t) = \langle 1 + 2t, 3 - t \rangle \quad \text{(line)}$$
$$\mathbf{s}(t) = \langle 2\cos(t), 2\sin(t) \rangle \quad \text{(circle of radius 2)}$$

- **Differentiation**: The derivative of $\mathbf{r}(t) = \langle x(t), y(t) \rangle$ is

$$\mathbf{r}'(t) = \langle x'(t), y'(t) \rangle.$$

- **Integration**: The indefinite integral of $\mathbf{r}(t) = \langle x(t), y(t) \rangle$ is

$$\int \mathbf{r}(t)\, dt = \left\langle \int x(t)\, dt, \int y(t)\, dt \right\rangle.$$

- **Two-Dimensional Motion**:

  1. Position vector: Position of moving object at time t.

$$\mathbf{r}(t) = \langle x(t), y(t) \rangle$$

     We usually draw the position vector $\mathbf{r}(t)$ from the origin to the object.

  2. Velocity vector: Rate of change of the position. The components correspond to the velocity in the x- and y-directions.

$$\mathbf{v}(t) = \mathbf{r}'(t) = \langle x'(t), y'(t) \rangle$$

     We usually draw the vector $\mathbf{v}(t)$ with the tail at the position of the object.

  3. Acceleration vector: Rate of change of the velocity. The components correspond to the acceleration in the x- and y-directions.

$$\mathbf{a}(t) = \mathbf{v}'(t) = \langle x''(t), y''(t) \rangle$$

     We usually draw the vector $\mathbf{a}(t)$ with the tail at the position of the object.

  4. Speed:

$$s(t) = |\mathbf{v}(t)|$$

- **Graphs of Functions**:

1. Tangent vector to the graph of $y = f(x)$ at $x = a$.
   Find $f'(x)$, then create the vector $\langle 1, f'(a)\rangle$ or $\langle \Delta x, f'(a)\Delta x\rangle$, for some choice of Δx.

   Example: Find a tangent vector to the graph of $f(x) = x^2 - x$ at $x = 3$.
   $f'(x) = 2x - 1$, so $f'(3) = 5$.
   A tangent vector to the graph $y = f(x)$ at $x = 3$ is $\langle 1, 5\rangle$.

2. Normal vector to the graph of $y = f(x)$ at $x = a$.

   Example: Find a normal vector to the graph of $f(x) = x^2 - x$ at $x = 3$.
   From the last example, the slope of the perpendicular vector is $-\frac{1}{5}$.
   A normal vector at $x = 3$ is $\langle 1, -\frac{1}{5}\rangle$ or $\langle 5, -1\rangle$.

